The $4+4+4+1$ Triality Cone and Its Dynamical Closure: Quaternionic Rank-Three Geometry, an Unlocked S^8, and Explicit Gauge Dynamics

Research continuation note

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Abstract

The previously established complex rank-three Hermitian object has real dimension nine and a Peirce decomposition $4+4+1$. This note determines what happens when one asks for three four-real-dimensional sectors coupled by an exact order-three symmetry rather than adjoining another mirrored summand.

There are three different, nested answers. The literal $4+4+4+1$ object is the equal-diagonal slice of the quaternionic Jordan algebra $\mathrm{Herm}_3(\mathbb{H})$:

$$\mathcal{T}_{13} = \mathbb{R} \oplus \mathbb{H}^3.$$

Its Moore determinant is

$$\mathcal{N}_{13}(\lambda; q_1, q_2, q_3) = \lambda^3 - \lambda \sum_i |q_i|^2 + 2\,\mathrm{Re}(q_1 q_2 q_3).$$

The last term is a genuine closed triangle interaction. The compactified determinant boundary is S^{12}, and the cyclic permutation of the three quaternionic edges is exact. More generally, the equal-diagonal rank-three slices over $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ have compactified boundaries S^3, S^6, S^{12}, S^{24}.

The thirteen-dimensional slice is not closed under Jordan multiplication or under the unrestricted determinant gradient. Its Jordan closure is the full fifteen-dimensional algebra $\mathrm{Herm}_3(\mathbb{H})$; fixing trace gives the fourteen-dimensional shape body. Relative to a chosen complex subalgebra, there is an exact decomposition

$$\mathrm{Herm}_3(\mathbb{H}) \cong \mathrm{Herm}_3(\mathbb{C}) \oplus \mathbb{C}^3, \qquad \det_{\mathbb{H}}(A, b) = \det_{\mathbb{C}} A - b^* A b.$$

Thus the old traceless eight-dimensional carrier embeds linearly in a fourteen-dimensional carrier as $\mathrm{Herm}_3^0(\mathbb{C}) \subset \mathrm{Herm}_3^0(\mathbb{H})$, giving a standard $S^8 \subset S^{14}$ with six transverse colour-triplet directions. The quaternionic positive cone projects onto the complex cone with fibres B^6, B^2, and a point over the rank-three, rank-two, and lower-rank strata. Its complex rank-one focal manifold $\mathbb{CP}^2$ opens into $\mathbb{HP}^2$; the retained SU(3) action has quotient interval with endpoint orbits $\mathbb{CP}^2$ and S^5.

Using the earlier A_2 observation matrix, the full fifteen-dimensional cubic decomposes as a Lorentz vector, two Weyl spinors, one real scalar, and one complex scalar. Removing the complex scalar gives a second thirteen-dimensional $4+4+4+1$ slice containing the old nine-dimensional object, but the removed scalar is precisely the term that directly mixes the two spinors. Finally, we construct a globally well-posed triality mechanics, a local $\mathrm{Sp}(1)^3$ triangle gauge theory, and an SU(3) Yang–Mills–scalar theory on the fourteen/fifteen-dimensional completion. The latter has an exact first-order vacuum transition whose broken vacuum manifold is $\mathbb{HP}^2$ and whose gauge quotient is the same internal interval. These are exact classical dynamical theories; no claim about pure quantum Yang–Mills or its mass gap is made.

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1 Quaternionic rank-three determinant

Fix quaternionic units i, j, k with $ij = k$. For $q \in \mathbb{H}$, let $\bar{q}$ denote quaternionic conjugation and $|q|^2 = q\bar{q}$. Put

$$Q(a; q) = \begin{pmatrix} a_1 & q_3 & \bar{q}_2 \\ \bar{q}_3 & a_2 & q_1 \\ q_2 & \bar{q}_1 & a_3 \end{pmatrix}, \qquad a_i \in \mathbb{R}, \quad q_i \in \mathbb{H}. \tag{1}$$

We use the explicit Moore determinant

$$\det_{\mathbb{H}} Q = a_1 a_2 a_3 - a_1|q_1|^2 - a_2|q_2|^2 - a_3|q_3|^2 + 2\,\mathrm{Re}(q_1 q_2 q_3). \tag{2}$$

The real part of a triple quaternionic product is cyclic, so the last term is unambiguous under cyclic relabelling. The positive cone

$$\mathrm{Herm}_3(\mathbb{H})_{\succeq 0}$$

is the cone of quaternionic Hermitian matrices with nonnegative real eigenvalues.

Definition 1.1 (Equal-diagonal triality slice)**.** Set

$$\mathcal{T}_{13} = \{Q(\lambda, \lambda, \lambda; q_1, q_2, q_3) : \lambda \in \mathbb{R},\ q_i \in \mathbb{H}\} \cong \mathbb{R} \oplus \mathbb{H}^3. \tag{3}$$

Its real dimension is $1 + 3 \cdot 4 = 13$.

Theorem 1.2 (The forced $4+4+4+1$ cubic)**.** *On $\mathcal{T}_{13}$, the determinant is*

$$\boxed{\mathcal{N}_{13}(\lambda; q_1, q_2, q_3) = \lambda^3 - \lambda(|q_1|^2 + |q_2|^2 + |q_3|^2) + 2\,\mathrm{Re}(q_1 q_2 q_3).} \tag{4}$$

It is invariant under the cyclic permutation

$$(q_1, q_2, q_3) \longmapsto (q_2, q_3, q_1)$$

and under the vertex gauge action

$$(s_1, s_2, s_3) : \begin{cases} q_1 \mapsto s_2 q_1 s_3^{-1}, \\ q_2 \mapsto s_3 q_2 s_1^{-1}, \\ q_3 \mapsto s_1 q_3 s_2^{-1}, \end{cases} \qquad s_i \in \mathrm{Sp}(1). \tag{5}$$

The common central element $(-1,-1,-1)$ acts trivially. Hence the effective compact gauge group is

$$K_\triangle = \mathrm{Sp}(1)^3/\{\pm(1,1,1)\}.$$

The three four-dimensional summands are therefore edges of one triangle rather than three independent copies. Their first joint invariant occurs in the closed product $\mathrm{Re}(q_1 q_2 q_3)$.

Proof. Substituting $a_1 = a_2 = a_3 = \lambda$ in (2) gives (4). Under (5), the product transforms as

$$q_1 q_2 q_3 \longmapsto s_2 (q_1 q_2 q_3) s_2^{-1},$$

whose real part is unchanged. Quaternionic norms are unchanged as well. Cyclic invariance follows from cyclicity of the real part. □

Proposition 1.3 (Positivity in triality coordinates)**.** *An element of $\mathcal{T}_{13}$ is positive semidefinite if and only if*

$$\lambda \geq 0, \qquad |q_i|^2 \leq \lambda^2 \quad (i = 1, 2, 3), \qquad \mathcal{N}_{13}(\lambda; q_1, q_2, q_3) \geq 0. \tag{6}$$

If $\lambda = 0$, these conditions force $q_1 = q_2 = q_3 = 0$.

Proof. The displayed quantities are exactly the Moore determinants of all principal 1×1, 2×2, and 3×3 submatrices, so they are necessary. For $\lambda > 0$, take the quaternionic Schur complement of the first diagonal entry. It is a 2×2 quaternionic Hermitian matrix whose two diagonal entries are $\lambda - |q_3|^2/\lambda$ and $\lambda - |q_2|^2/\lambda$, and whose Moore determinant is $\mathcal{N}_{13}/\lambda$. The 2×2 positivity criterion therefore proves sufficiency. The case $\lambda = 0$ follows from the three quadratic inequalities. □

Proposition 1.4 (Generic orbit coordinates)**.** *On the open set $q_1 q_2 q_3 \neq 0$, the $K_\triangle$-orbit is determined by*

$$r_i = |q_i| > 0, \qquad \tau = \mathrm{Re}(q_1 q_2 q_3), \qquad -r_1 r_2 r_3 \leq \tau \leq r_1 r_2 r_3. \tag{7}$$

Equivalently, write

$$\tau = r_1 r_2 r_3 \cos\theta, \qquad 0 \leq \theta \leq \pi.$$

At fixed λ, r_1, r_2, r_3, the determinant boundary is the explicit phase equation

$$\cos\theta = \frac{\lambda(r_1^2 + r_2^2 + r_3^2) - \lambda^3}{2 r_1 r_2 r_3}, \tag{8}$$

whenever the right side lies in $[-1, 1]$.

Proof. Use (5) to make $q_1 = r_1$ and $q_2 = r_2$ positive real. The residual gauge group is common conjugation by one unit quaternion, acting on q_3. Quaternionic conjugacy classes are determined by norm and real part. Since $q_1 q_2 q_3 = r_1 r_2 q_3$ in this gauge, (7) is complete. Equation (8) is (4) with determinant zero. □

2 The sphere ladder forced by equal-diagonal rank three

Let $\mathbb{A}$ be one of the four normed real division algebras $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$, of real dimension $d = 1, 2, 4, 8$. Let $\mathcal{T}(\mathbb{A})$ be the equal-diagonal slice of the rank-three Hermitian Jordan algebra over $\mathbb{A}$. In the octonionic case, the real part of the associator vanishes, so the cubic formula remains well-defined.

Theorem 2.1 (Division-algebra boundary ladder)**.** *The real dimension of $\mathcal{T}(\mathbb{A})$ is $1 + 3d$. Let*

$$\mathcal{C}(\mathbb{A}) = \mathcal{T}(\mathbb{A}) \cap \mathrm{Herm}_3(\mathbb{A})_{\succeq 0}.$$

Then

$$\mathcal{C}(\mathbb{A})^+ \cong B^{1+3d}, \qquad (\partial\mathcal{C}(\mathbb{A}))^+ \cong S^{3d}. \tag{9}$$

Consequently the equal-diagonal ladder is

$\mathbb{A}$	$\dim_{\mathbb{R}} \mathbb{A}$	$\dim_{\mathbb{R}} \mathcal{T}(\mathbb{A})$	$(\partial\mathcal{C}(\mathbb{A}))^+$
$\mathbb{R}$	1	4	S^3
$\mathbb{C}$	2	7	S^6
$\mathbb{H}$	4	13	S^{12}
$\mathbb{O}$	8	25	S^{24}.

(10)

Every chosen inclusion of division algebras induces a determinant- and positivity-preserving linear inclusion of cones and therefore a based embedding of the displayed compactified boundaries.

Proof. A nonzero positive-semidefinite equal-diagonal matrix has $\lambda > 0$. Its intersection with the affine hyperplane $\lambda = 1$ is a compact convex body in $\mathbb{A}^3$: compactness follows from the principal-minor inequalities $|q_i| \le 1$, and the origin of $\mathbb{A}^3$ is an interior point. Every compact convex body with nonempty interior is homeomorphic to a closed ball by radial projection from an interior point, with boundary a sphere. Positive scaling reconstructs the cone. Thus the cone is homeomorphic to a closed half-space of dimension $1 + 3d$, and (9) follows. Entrywise algebra inclusions preserve the determinant formula and positivity. □

Remark 2.2 (What the ladder does and does not identify)*.* The complex rung supplies a canonical S^6 as the compactified determinant boundary of a seven-dimensional real cone. The quaternionic and octonionic rungs explain the previously recurring dimensions S^{12} and S^{24}. These are exact sphere carriers. An identification with any separately constructed torus-fibred complex threefold would still require a map preserving its period law and singular fillings.

3 Why the thirteen-dimensional slice generates fifteen dimensions

Let

$$\mathcal{J}_{15} = \mathrm{Herm}_3(\mathbb{H}).$$

This is the rank-three quaternionic Euclidean Jordan algebra [1, 3]; as a real vector space it has dimension $3 + 3 \cdot 4 = 15$.

Theorem 3.1 (Jordan closure)**.** *The smallest Jordan subalgebra of $\mathcal{J}_{15}$ containing $\mathcal{T}_{13}$ is all of $\mathcal{J}_{15}$.*

Proof. The slice $\mathcal{T}_{13}$ contains the identity and every off-diagonal Hermitian edge matrix. Let $E_1(q)$ be supported in the $(2, 3)$ and $(3, 2)$ positions. For $|q| = 1$,

$$E_1(q)^2 = \mathrm{diag}(0, 1, 1).$$

Likewise the other two edges square to $\mathrm{diag}(1,0,1)$ and $\mathrm{diag}(1,1,0)$. Their linear combinations give all three diagonal matrix units, for example

$$E_{11} = \frac{1}{2}\big(\mathrm{diag}(1,0,1) + \mathrm{diag}(1,1,0) - \mathrm{diag}(0,1,1)\big).$$

Together with the already present quaternionic off-diagonal entries, these span $\mathrm{Herm}_3(\mathbb{H})$. Since $X \circ X = X^2$, every square used above belongs to the Jordan closure. □

Write

$$a_i = \lambda + \eta_i, \qquad \eta_1 + \eta_2 + \eta_3 = 0. \tag{11}$$

The two-dimensional vector η is the diagonal A_2 Cartan anisotropy omitted by the equal-diagonal slice.

Proposition 3.2 (Exact anisotropy coupling)**.** *With $r_i^2 = |q_i|^2$, the full quaternionic determinant is*

$$\begin{aligned}\det_{\mathbb{H}} Q = \mathcal{N}_{13}(\lambda; q_1, q_2, q_3) - \frac{\lambda}{2}(\eta_1^2 + \eta_2^2 + \eta_3^2) + \eta_1\eta_2\eta_3 \\ - (\eta_1 r_1^2 + \eta_2 r_2^2 + \eta_3 r_3^2).\end{aligned} \tag{12}$$

At $\eta = 0$, its derivative in a trace-free diagonal direction $\delta\eta$ is

$$D_\eta \det[\delta\eta] = -\sum_{i=1}^{3} r_i^2 \delta\eta_i. \tag{13}$$

The normal gradient vanishes for all $\delta\eta_1 + \delta\eta_2 + \delta\eta_3 = 0$ if and only if

$$|q_1| = |q_2| = |q_3|. \tag{14}$$

Thus the thirteen-dimensional slice is generically not invariant under unrestricted determinant-gradient dynamics. The two missing real coordinates are sourced precisely by imbalance among the three four-dimensional sectors.

Proof. Expand the product $\prod_i(\lambda + \eta_i)$ and use

$$\eta_1\eta_2 + \eta_2\eta_3 + \eta_3\eta_1 = -\frac{1}{2}\sum_i \eta_i^2.$$

This gives (12). Differentiating at zero gives (13). A vector is orthogonal to the trace-free plane precisely when its three coordinates are equal. □

Corollary 3.3 (The roles of dimensions thirteen, fourteen, and fifteen)**.** *The three dimensions have distinct exact meanings.*

1. 13*: $\mathcal{T}_{13} = \mathbb{R} \oplus \mathbb{H}^3$ is the triality-symmetric ansatz.*
2. 14*: the fixed-trace section*

   $$\{Q \in \mathrm{Herm}_3(\mathbb{H})_{\succeq 0} : \mathrm{tr}\, Q = \tau\}, \qquad \tau > 0,$$

   is a compact convex 14-ball whose boundary is S^{13}. It contains all shape and anisotropy variables but removes overall radial scale.
3. 15*: $\mathrm{Herm}_3(\mathbb{H})$ is the unconstrained Jordan-closed linear space and its positive cone has compactified determinant boundary S^{14}.*

4 Exact quaternionic enlargement of the old nine-dimensional object

Fix the complex subalgebra $\mathbb{C}_i = \mathbb{R} \oplus \mathbb{R}i \subset \mathbb{H}$. Every quaternion is uniquely $z + wj$ with $z, w \in \mathbb{C}_i$.

For $A \in \mathrm{Herm}_3(\mathbb{C}_i)$ and $b = (b_1, b_2, b_3)^{\mathsf{T}} \in \mathbb{C}_i^3$, define

$$B(b) = \begin{pmatrix} 0 & \bar{b}_3 & -\bar{b}_2 \\ -\bar{b}_3 & 0 & \bar{b}_1 \\ \bar{b}_2 & -\bar{b}_1 & 0 \end{pmatrix} \tag{15}$$

and

$$\Theta(A, b) = A + B(b)j \in \mathrm{Herm}_3(\mathbb{H}). \tag{16}$$

Theorem 4.1 (Complex-plus-triplet decomposition)**.** *The map*

$$\Theta : \mathrm{Herm}_3(\mathbb{C}_i) \oplus \mathbb{C}_i^3 \longrightarrow \mathrm{Herm}_3(\mathbb{H})$$

is a real-linear isomorphism, and

$$\boxed{\det_{\mathbb{H}} \Theta(A, b) = \det_{\mathbb{C}} A - b^* A b.} \tag{17}$$

For $U \in \mathrm{SU}(3)$,

$$\Theta(A, b) \longmapsto \Theta(UAU^*, Ub) \tag{18}$$

is exactly quaternionic congruence by the complex matrix U and preserves the determinant and positive cone.

Proof. The skew matrix $B(b)$ records the three components orthogonal to $\mathbb{C}_i$, so uniqueness is immediate. Write the three quaternionic off-diagonal entries as $q_r = z_r + \bar{b}_r j$. Expanding (2) with

$$(z + wj)(z' + w'j) = zz' - w\bar{w}' + (zw' + w\bar{z}')j$$

gives (17); the accompanying symbolic replay performs the full real-coordinate expansion. The identity

$$UB(b)U^{\mathsf{T}} = B(Ub), \qquad \det U = 1,$$

proves (18). Quaternionic unitary congruence preserves positivity. □

Corollary 4.2 (The fourteen-dimensional S^8 unlock)**.** *Taking traces in* (16) *gives*

$$\mathrm{Herm}_3^0(\mathbb{H}) \cong \mathrm{Herm}_3^0(\mathbb{C}_i) \oplus \mathbb{C}_i^3 \cong \mathbb{R}^8 \oplus \mathbb{R}^6. \tag{19}$$

Therefore the one-point compactification of the old traceless carrier embeds as the standard unknotted linear sphere

$$\boxed{S^8 \hookrightarrow S^{14}.} \tag{20}$$

Its normal representation is the complex colour triplet $\mathbb{C}^3$. The inclusion is $\mathrm{SU}(3)$*-equivariant under conjugation on the first summand and the fundamental action on the second.*

Proof. The trace is carried entirely by A. The dimensions are eight and six. One-point compactification of the linear inclusion $\mathbb{R}^8 \subset \mathbb{R}^{14}$ gives (20). □

Theorem 4.3 (Commuting Gram-boundary unlock)**.** *For $\mathbb{A} = \mathbb{C}_i$ or $\mathbb{H}$, define*

$$\Phi_{\mathbb{A}} : \mathrm{Herm}_3^0(\mathbb{A}) \longrightarrow \partial\,\mathrm{Herm}_3(\mathbb{A})_{\succeq 0}, \qquad \Phi_{\mathbb{A}}(K) = K - \lambda_{\min}(K) I_3. \tag{21}$$

Then $\Phi_{\mathbb{A}}$ is a positively homogeneous equivariant homeomorphism with inverse

$$\Psi_{\mathbb{A}}(Q) = Q - \frac{\operatorname{tr} Q}{3} I_3. \tag{22}$$

The complex/quaternionic inclusions form the commuting square

$$\begin{array}{ccc} \mathrm{Herm}_3^0(\mathbb{C}_i) & \hookrightarrow & \mathrm{Herm}_3^0(\mathbb{H}) \\ \downarrow{\scriptstyle \Phi_{\mathbb{C}_i}} & & \downarrow{\scriptstyle \Phi_{\mathbb{H}}} \\ \partial\,\mathrm{Herm}_3(\mathbb{C}_i)_{\succeq 0} & \hookrightarrow & \partial\,\mathrm{Herm}_3(\mathbb{H})_{\succeq 0}. \end{array} \tag{23}$$

After one-point compactification this is the exact boundary embedding $S^8 \hookrightarrow S^{14}$, not merely a dimension match.

Proof. For trace-free K, subtracting its least real eigenvalue makes all eigenvalues nonnegative and at least one zero. Conversely, if $Q \succeq 0$ is singular, then the least eigenvalue of $Q - (\operatorname{tr} Q/3) I_3$ is $-\operatorname{tr} Q/3$. The two formulas are therefore inverse, and continuity follows from continuity of ordered quaternionic Hermitian eigenvalues. A complex Hermitian matrix has the same eigenvalues when viewed as quaternionic Hermitian, so the square commutes. Equivariance follows from unitary congruence invariance of the spectrum. □

Theorem 4.4 (Minimality of the full-colour unlock)**.** *Under the decomposition of Theorem 4.1, the standard complex representation $\mathbb{C}^3$ is irreducible as a real $\mathrm{SU}(3)$-representation. Consequently:*

1. *the smallest real $\mathrm{SU}(3)$-invariant linear subspace of $\mathrm{Herm}_3^0(\mathbb{H})$ containing $\mathrm{Herm}_3^0(\mathbb{C}_i)$ and one nonzero transverse vector is all of*

   $$\mathrm{Herm}_3^0(\mathbb{H}) = \mathrm{Herm}_3^0(\mathbb{C}_i) \oplus \mathbb{C}_i^3,$$

   of real dimension fourteen;

2. *the smallest real $\mathrm{SU}(3)$-invariant linear subspace of $\mathrm{Herm}_3(\mathbb{H})$ containing $\mathrm{Herm}_3(\mathbb{C}_i)$ and one nonzero transverse vector is all of $\mathrm{Herm}_3(\mathbb{H})$, of real dimension fifteen.*

In particular, no thirteen-dimensional linear completion of the old nine-dimensional object can both add a nonzero quaternionic direction and retain full $\mathrm{SU}(3)$ invariance. Every thirteen-dimensional completion below is therefore a specified symmetry reduction, and the inclusions into the fourteen- and fifteen-dimensional carriers are the exact morphisms relating those objects.

Proof. The group $\mathrm{SU}(3)$ acts transitively on the unit sphere in $\mathbb{C}^3$. Hence a nonzero invariant real subspace containing one transverse vector contains its entire unit-sphere orbit and therefore spans all of $\mathbb{C}^3$ over $\mathbb{R}$. The two claims now follow from the direct sums in (19) and Theorem 4.1. □

4.1 The focal $\mathbb{CP}^2$ opens into $\mathbb{HP}^2$

Let

$$\mathcal{P}_{\mathbb{C}} = \{P \in \mathrm{Herm}_3(\mathbb{C}_i) : P^2 = P,\ \operatorname{tr} P = 1\}, \qquad \mathcal{P}_{\mathbb{H}} = \{P \in \mathrm{Herm}_3(\mathbb{H}) : P^2 = P,\ \operatorname{tr} P = 1\}.$$

These are respectively the complex and quaternionic rank-one projector manifolds $\mathbb{CP}^2$ and $\mathbb{HP}^2$, with the first included in the second.

Theorem 4.5 (SU(3) orbit interval on $\mathbb{HP}^2$)**.** *Represent a quaternionic line by a unit vector*

$$v = z + wj \in \mathbb{H}^3, \qquad z, w \in \mathbb{C}^3_i, \qquad \|z\|^2 + \|w\|^2 = 1,$$

and put $Z = (z\ w) \in \mathrm{Mat}_{3\times 2}(\mathbb{C}_i)$*. The function*

$$\delta([v]) = 4\det(Z^*Z) \tag{24}$$

is well defined on $\mathbb{HP}^2$*, is* SU(3)*-invariant, has range* $[0,1]$*, and induces a homeomorphism*

$$\boxed{\mathbb{HP}^2/\operatorname{SU}(3) \cong [0,1].} \tag{25}$$

The orbit types are

δ	*orbit*	*stabilizer in* SU(3)
0	$\mathbb{CP}^2$	$S(U(1)\times U(2))$
$0<\delta<1$	$\operatorname{SU}(3)/U(1)_{1,-1,0}$	$U(1)_{1,-1,0}$
1	$S^5 = \operatorname{SU}(3)/\operatorname{SU}(2)$	SU(2).

(26)

Here

$$U(1)_{1,-1,0} = \{\mathrm{diag}(e^{i\theta}, e^{-i\theta}, 1) : \theta \in \mathbb{R}\}.$$

Consequently $\mathbb{HP}^2$ *has the explicit double disk-bundle description*

$$\boxed{\mathbb{HP}^2 \cong \operatorname{SU}(3) \times_{S(U(1)\times U(2))} D^4 \ \cup_{\operatorname{SU}(3)/U(1)_{1,-1,0}} \ \operatorname{SU}(3) \times_{\operatorname{SU}(2)} D^3.} \tag{27}$$

The old complex focal manifold is exactly the endpoint $\delta = 0$*. This recovers, in the present Gram coordinates, the standard cohomogeneity-one* SU(3) *action on* $\mathbb{HP}^2$ *[4].*

Proof. Right multiplication of $v = z + wj$ by a unit quaternion $a + bj$ sends

$$(z\ w) \longmapsto (z\ w)\begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix},$$

and the displayed matrix lies in SU(2). Left SU(3) multiplication and right SU(2) multiplication commute. The left SU(3) orbit of a 3×2 complex matrix is classified by its Gram matrix Z^*Z; the right SU(2) action conjugates that Gram matrix. Because its trace is one, its conjugacy class is determined by its two eigenvalues. Thus $4\det(Z^*Z) \in [0,1]$ is a complete orbit parameter.

For eigenvalues $\lambda_1 \geq \lambda_2 \geq 0$ with sum one, a representative is

$$Z_\lambda = \begin{pmatrix} \sqrt{\lambda_1} & 0 \\ 0 & \sqrt{\lambda_2} \\ 0 & 0 \end{pmatrix}.$$

If $\lambda_2 = 0$, the stabilizer is $S(U(1)\times U(2))$ and the orbit is $\mathbb{CP}^2$. If $\lambda_1 > \lambda_2 > 0$, compatibility with a right SU(2) transformation leaves precisely $U(1)_{1,-1,0}$. If $\lambda_1 = \lambda_2 = 1/2$, the stabilizer is the upper-left SU(2) and the orbit is S^5. The normal spheres are $S(U(1)\times U(2))/U(1)_{1,-1,0} \cong S^3$ and $\operatorname{SU}(2)/U(1)_{1,-1,0} \cong S^2$, which gives (27). □

Proposition 4.6 (The orbit coordinate is the transverse triplet norm)**.** *Let $P = vv^*$ be the projector associated with the normalized vector in Theorem 4.5, and write $P = \Theta(A_P, b_P)$. Then*

$$\boxed{b_P^* b_P = \|z \wedge w\|^2 = \det(Z^*Z) = \frac{\delta([v])}{4}.} \tag{28}$$

For $Q = x(3P - I_3)$, its triplet component is $b_Q = 3xb_P$, so

$$\boxed{b_Q^* b_Q = \frac{9x^2}{4}\delta([v]).} \tag{29}$$

Thus the ordinary SU(3)*-invariant quadratic b^*b is an exact coordinate along the internal orbit interval.*

Proof. Writing $v_r = z_r + w_r j$, the coefficient of j in $v_r\overline{v_s}$ is $w_r z_s - z_r w_s$. The three components of b_P are, up to the fixed signs in (15), the three Pluecker coordinates of the pair (z, w). The complex Cauchy–Binet identity gives

$$\sum_{r<s} |z_r w_s - z_s w_r|^2 = \det(Z^*Z).$$

Linearity of Θ gives the second formula. □

Proposition 4.7 (A boundary path leaving the complex S^8)**.** *Let $j \perp \mathbb{C}_i$ be the fixed quaternionic unit. For $t \in \mathbb{R}$, set*

$$Q_t = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & tj \\ 0 & -tj & t^2 \end{pmatrix}. \tag{30}$$

Then $Q_t \succeq 0$ and $\operatorname{rank} Q_t = 2$ for every t. At $t = 0$ it lies in the complex rank-defect cone; for $t \neq 0$ it leaves $\mathrm{Herm}_3(\mathbb{C}_i)$ while remaining on the quaternionic determinant boundary.

Proof. The lower block is vv^* for $v = (1, -tj)^{\mathsf{T}}$, hence is positive semidefinite of rank one. The first diagonal entry contributes one further rank. The off-diagonal entry tj is not in $\mathbb{C}_i$ when $t \neq 0$. □

4.2 The positive projection and its collapsing fibres

Let

$$\pi_{\mathbb{C}} : \mathrm{Herm}_3(\mathbb{H})_{\succeq 0} \longrightarrow \mathrm{Herm}_3(\mathbb{C}_i)_{\succeq 0}, \qquad \pi_{\mathbb{C}}(\Theta(A, b)) = A.$$

It is the average of the identity with entrywise conjugation by i, so it is positive and SU(3)-equivariant.

Theorem 4.8 (Exact fibre topology over the complex cone)**.** *The map $\pi_{\mathbb{C}}$ is proper and surjective. Its fibres are as follows.*

1. *If $A > 0$, then*

$$\pi_{\mathbb{C}}^{-1}(A) = \{\Theta(A, b) : b^* A b \leq \det A\}, \tag{31}$$

   a closed real six-ball. Equality is the rank-two boundary sphere S^5.

2. *If $A \succeq 0$ has rank two and positive eigenvalues a_1, a_2, then*

$$\pi_{\mathbb{C}}^{-1}(A) = \{\Theta(A, b) : b \in \ker A,\ |b|^2 \leq a_1 a_2\}, \tag{32}$$

   a closed real two-ball. Its interior has quaternionic rank two and its boundary has rank one.

3. If $\operatorname{rank} A \le 1$*, the fibre is the single point* $b = 0$*.*

Thus the full quaternionic boundary maps to the complex cone as an S^5*-bundle over the interior, a* B^2*-bundle over the generic complex boundary, and a point over the lower-rank locus. The old* S^8 *is not discarded; it is the* $b = 0$ *complex boundary inside this larger stratified boundary.*

Proof. For $A > 0$, quaternionic congruence by $A^{-1/2}$ transforms $\Theta(A, b)$ into $\Theta(I, \beta)$ with

$$\beta = (\det A)^{-1/2} A^{1/2} b.$$

An SU(3) matrix sends β to $(r, 0, 0)$. The resulting quaternionic Hermitian matrix has eigenvalues $1, 1 - r, 1 + r$. Hence positivity is equivalent to $r^2 = b^* A b / \det A \le 1$, proving (31).

For rank two, diagonalize A unitarily to $\operatorname{diag}(a_1, a_2, 0)$. Positivity of the three 2×2 principal submatrices forces the first two components of b to vanish and gives $|b_3|^2 \le a_1 a_2$. Rank one forces all three components to vanish. Properness follows because $\operatorname{tr} \Theta(A, b) = \operatorname{tr} A$ and a positive-semidefinite matrix has norm bounded by its trace on compact trace ranges. □

Corollary 4.9 (Cone compactifications)**.** *The positive cones and their determinant boundaries satisfy*

$$\operatorname{Herm}_3(\mathbb{C}_i)^+_{\succeq 0} \cong B^9, \quad (\partial \operatorname{Herm}_3(\mathbb{C}_i)_{\succeq 0})^+ \cong S^8, \tag{33}$$

$$\operatorname{Herm}_3(\mathbb{H})^+_{\succeq 0} \cong B^{15}, \quad (\partial \operatorname{Herm}_3(\mathbb{H})_{\succeq 0})^+ \cong S^{14}. \tag{34}$$

The inclusion extends to $S^8 \hookrightarrow S^{14}$*. The boundary restriction of* $\pi_{\mathbb{C}}$ *does not land in the old* S^8*, because the new* S^5 *boundary fibres lie over positive-definite* A*.*

5 Two different thirteen-dimensional slices inside the same closure

The equal-diagonal $\mathcal{T}_{13}$ is the manifestly triality-symmetric slice. A second thirteen-dimensional slice is forced if one instead insists on containing the complete old nine-dimensional Hermitian object and adding exactly one four-dimensional Lorentz-spinor sector. Their relation is through the common fifteen-dimensional closure, not through an identification:

$$\begin{array}{ccccc} \mathcal{T}_{13} \cap \operatorname{Herm}_3(\mathbb{C}_i) = \mathcal{T}(\mathbb{C}_i) & \hookrightarrow & \mathcal{T}_{13} & \hookrightarrow & \operatorname{Herm}_3(\mathbb{H}) \\ \operatorname{Herm}_3(\mathbb{C}_i) & \hookrightarrow & \mathcal{L}_{13} & \hookrightarrow & \operatorname{Herm}_3(\mathbb{H}). \end{array} \tag{35}$$

The upper intersection has real dimension seven, whereas the lower intersection is the complete old nine-dimensional object. Thus the upper route preserves manifest three-edge triality and the lower route preserves the entire old carrier.

Retain the earlier matrices

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad u = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \qquad R_P = P^*(PP^*)^{-1}, \qquad X = (R_P \; u), \tag{36}$$

with $\det X = 1$. For $A \in \operatorname{Herm}_3(\mathbb{C})$ and $b \in \mathbb{C}^3$, define

$$X^{-1} A X^{-*} = \begin{pmatrix} G & z \\ z^* & \lambda \end{pmatrix}, \qquad X^* b = \begin{pmatrix} \chi \\ \rho \end{pmatrix}, \tag{37}$$

where $G \in \operatorname{Herm}_2(\mathbb{C})$, $z, \chi \in \mathbb{C}^2$, $\lambda \in \mathbb{R}$, and $\rho \in \mathbb{C}$.

Theorem 5.1 (Full $4+4+4+1+2$ cubic)**.** *In the coordinates* (37)*,*

$$\det_{\mathbb{H}} \Theta(A,b) = \lambda \det G - z^* \operatorname{adj}(G) z - \chi^* G \chi - 2\operatorname{Re}(\rho\, \chi^* z) - \lambda |\rho|^2. \tag{38}$$

For $g \in SL_2(\mathbb{C})$, put

$$T_P(g) = X(g \oplus 1)X^{-1}.$$

The determinant-preserving action

$$A \mapsto T_P(g) A T_P(g)^*, \qquad b \mapsto T_P(g)^{-*} b \tag{39}$$

becomes

$$G \mapsto gGg^*, \quad z \mapsto gz, \quad \chi \mapsto g^{-*}\chi, \quad \lambda \mapsto \lambda, \quad \rho \mapsto \rho. \tag{40}$$

Thus the full fifteen-dimensional space is a Lorentz vector, a left Weyl spinor, a dual Weyl spinor, a real scalar, and a complex scalar:

$$15 = 4 + 4 + 4 + 1 + 2.$$

Proof. Because $\det X = 1$,

$$\det A = \lambda \det G - z^* \operatorname{adj}(G) z.$$

Also $b = X^{-*}(\chi, \rho)^{\mathsf{T}}$, so

$$b^* A b = (\chi^*, \bar{\rho}) \begin{pmatrix} G & z \\ z^* & \lambda \end{pmatrix} \begin{pmatrix} \chi \\ \rho \end{pmatrix}.$$

Expanding and using (17) gives (38). The transformation law follows by conjugating with X. □

Corollary 5.2 (Lorentz $4+4+4+1$ slice)**.** *The condition $\rho = 0$, equivalently $u^* b = 0$, defines an invariant real thirteen-dimensional subspace*

$$\mathcal{L}_{13} = \{(G, z, \chi, \lambda) : G \in \operatorname{Herm}_2(\mathbb{C}),\ z, \chi \in \mathbb{C}^2,\ \lambda \in \mathbb{R}\} \tag{41}$$

with cubic

$$\boxed{\mathcal{N}_{\mathcal{L}}(G, z, \chi, \lambda) = \lambda \det G - z^* \operatorname{adj}(G) z - \chi^* G \chi.} \tag{42}$$

It contains the complete old nine-dimensional object as the subspace $\chi = 0$. Its positive cone has one-point compactification B^{13} and compactified determinant boundary S^{12}; the old compactified S^8 embeds properly in that S^{12}.

Remark 5.3 (Why the missing two dimensions matter)*.* The restriction (42) contains no term linear in both spinors. In the full cubic, the omitted complex scalar supplies precisely

$$-2\operatorname{Re}(\rho \chi^* z).$$

Consequently $\rho \neq 0$ permits one spinor to source the other. The direct continuation of the old S^8 therefore becomes fully mixed only in fifteen dimensions; the thirteen-dimensional Lorentz slice is a consistent symmetry reduction, not the complete dynamical closure.

Proposition 5.4 (The exact order-three matrix)**.** *Let*

$$\Pi = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \in \mathrm{SU}(3)$$

be the cyclic colour permutation. Then

$$P\Pi R_P = C_1 := \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}, \qquad C_1^3 = I_2, \qquad T_P(C_1) = \Pi. \tag{43}$$

Thus the three-element colour cycle acts on the Lorentz blocks by

$$G \mapsto C_1 G C_1^*, \qquad z \mapsto C_1 z, \qquad \chi \mapsto C_1^{-*}\chi,$$

with λ and ρ fixed.

6 A globally well-posed triality mechanics

Put

$$\mathscr{R}^2 = \lambda^2 + |q_1|^2 + |q_2|^2 + |q_3|^2. \tag{44}$$

Fix constants

$$\mu^2 \geq 0, \qquad \gamma > 0, \qquad \kappa \in \mathbb{R}.$$

Define

$$V_{13} = \frac{\mu^2}{2}\mathscr{R}^2 + \frac{\gamma}{4}\mathscr{R}^4 - \kappa \mathcal{N}_{13}. \tag{45}$$

Theorem 6.1 (Nonlinear triality dynamics)**.** *The Lagrangian*

$$L_{13} = \frac{1}{2}\dot{\lambda}^2 + \frac{1}{2}\sum_{i=1}^{3} |\dot{q}_i|^2 - V_{13} \tag{46}$$

is invariant under $K_\triangle \rtimes C_3$. Its equations are

$$\ddot{\lambda} + (\mu^2 + \gamma\mathscr{R}^2)\lambda - \kappa\left(3\lambda^2 - \sum_i |q_i|^2\right) = 0, \tag{47}$$

$$\ddot{q}_1 + (\mu^2 + \gamma\mathscr{R}^2)q_1 - 2\kappa(\bar{q}_3\bar{q}_2 - \lambda q_1) = 0, \tag{48}$$

$$\ddot{q}_2 + (\mu^2 + \gamma\mathscr{R}^2)q_2 - 2\kappa(\bar{q}_1\bar{q}_3 - \lambda q_2) = 0, \tag{49}$$

$$\ddot{q}_3 + (\mu^2 + \gamma\mathscr{R}^2)q_3 - 2\kappa(\bar{q}_2\bar{q}_1 - \lambda q_3) = 0. \tag{50}$$

The energy

$$E_{13} = \frac{1}{2}\dot{\lambda}^2 + \frac{1}{2}\sum_i |\dot{q}_i|^2 + V_{13} \tag{51}$$

is conserved, V_{13} is bounded below and coercive, and every initial condition has a unique solution for all real time.

Proof. The invariance follows from Theorem 1.2. With the Euclidean quaternionic inner product, differentiation of (4) gives

$$\nabla_{q_1}\mathcal{N}_{13} = 2(\bar{q}_3\bar{q}_2 - \lambda q_1),$$
$$\nabla_{q_2}\mathcal{N}_{13} = 2(\bar{q}_1\bar{q}_3 - \lambda q_2),$$
$$\nabla_{q_3}\mathcal{N}_{13} = 2(\bar{q}_2\bar{q}_1 - \lambda q_3),$$

and $\partial_\lambda \mathcal{N}_{13} = 3\lambda^2 - \sum_i |q_i|^2$. This proves the equations and energy conservation.

There is a constant C with $|\mathcal{N}_{13}| \le C\mathscr{R}^3$: use $2|q_1q_2q_3| \le 2(\sum_i |q_i|^2/3)^{3/2}$ and the triangle inequality. Therefore the positive quartic term dominates the cubic term at infinity. A fixed energy bounds positions and velocities, preventing finite-time escape; standard smooth ODE theory gives global existence and uniqueness. □

Corollary 6.2 (The third sector is dynamically generated)**.** *Suppose at some time t_0,*

$$q_3(t_0) = \dot{q}_3(t_0) = 0, \qquad q_1(t_0)q_2(t_0) \neq 0, \qquad \kappa \neq 0.$$

Then

$$\boxed{\ddot{q}_3(t_0) = 2\kappa\, \bar{q}_2(t_0)\bar{q}_1(t_0) \neq 0.} \tag{52}$$

Hence a two-edge configuration does not remain a mirrored two-sector system: the closed cubic interaction immediately creates the third four-dimensional sector.

7 Local gauge-field lifts

7.1 A triangle gauge theory with thirteen scalar components

Let spacetime be four-dimensional Minkowski space. At three vertices take gauge fields $A_\mu^{(r)} \in \operatorname{Im}\mathbb{H} \cong \mathfrak{su}(2)$. Orient q_1 from vertex 3 to 2, q_2 from 1 to 3, and q_3 from 2 to 1, consistently with (5). Define

$$\begin{aligned} D_\mu q_1 &= \partial_\mu q_1 + A_\mu^{(2)} q_1 - q_1 A_\mu^{(3)}, \\ D_\mu q_2 &= \partial_\mu q_2 + A_\mu^{(3)} q_2 - q_2 A_\mu^{(1)}, \\ D_\mu q_3 &= \partial_\mu q_3 + A_\mu^{(1)} q_3 - q_3 A_\mu^{(2)}. \end{aligned} \tag{53}$$

Theorem 7.1 (Power-counting renormalizable classical triangle model)**.** *The density*

$$\mathcal{L}_\triangle = -\frac{1}{4g^2}\sum_{r=1}^{3}\langle F_{\mu\nu}^{(r)}, F^{(r)\mu\nu}\rangle + \frac{1}{2}\partial_\mu\lambda\partial^\mu\lambda + \frac{1}{2}\sum_{i=1}^{3}|D_\mu q_i|^2 - V_{13} \tag{54}$$

is invariant under local $\mathrm{Sp}(1)^3/\{\pm 1\}$ gauge transformations and the global cyclic permutation of the three vertices. In four dimensions,

$$[A_\mu] = [q_i] = [\lambda] = 1, \quad [g] = [\gamma] = 0, \quad [\mu^2] = 2, \quad [\kappa] = 1.$$

Every term has engineering dimension four. Because $\gamma > 0$, the scalar potential is bounded below.

Proof. The covariant derivatives transform by the same left-right rule as their edge fields. The loop product $q_1q_2q_3$ transforms by conjugation at one vertex, so its real part is gauge invariant. The remaining claims follow from the displayed dimensions and the coercivity estimate in Theorem 6.1. □

Remark 7.2. This is a genuine local classical gauge theory with a three-node symmetry and a closed-loop cubic interaction. Its gauge group is $\mathrm{SU}(2)^3$ modulo a central kernel, not colour SU(3).

7.2 The SU(3)-equivariant fourteen/fifteen-dimensional model

The decomposition (17) gives a separate colour theory. Let

$$\Phi(x) \in \mathrm{Herm}_3^0(\mathbb{C}), \qquad b(x) \in \mathbb{C}^3,$$

so the scalar target is the fourteen-dimensional space in (19). Let $\mathcal{A}_\mu \in \mathfrak{su}(3)$ and

$$D_\mu \Phi = \partial_\mu \Phi + [\mathcal{A}_\mu, \Phi], \qquad D_\mu b = \partial_\mu b + \mathcal{A}_\mu b.$$

Put

$$\mathscr{S}^2 = \mathrm{tr}(\Phi^2) + 2b^* b, \qquad \mathcal{N}_{14}(\Phi, b) = \det \Phi - b^* \Phi b. \tag{55}$$

Theorem 7.3 (SU(3) dynamical unlock of the S^8 carrier)**.** *For constants $g_c > 0$, $m^2 \geq 0$, $\Gamma > 0$, and $\kappa \in \mathbb{R}$, the Lagrangian density*

$$\begin{aligned} \mathcal{L}_{14} = &- \frac{1}{2g_c^2} \mathrm{tr}(\mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu}) + \frac{1}{2} \mathrm{tr}(D_\mu \Phi D^\mu \Phi) + (D_\mu b)^* D^\mu b \\ &- \frac{m^2}{2} \mathscr{S}^2 - \frac{\Gamma}{4} \mathscr{S}^4 + \kappa \mathcal{N}_{14}(\Phi, b) \end{aligned} \tag{56}$$

is a local, power-counting renormalizable, SU(3) *gauge theory. The old eight-dimensional field space is the invariant locus $b = 0$; its one-point compactification is the embedded $S^8 \subset S^{14}$ of Theorem 4.2.*

At a constant background $(\Phi_0, 0)$, the quadratic operator on the six transverse real modes b has Hermitian mass matrix

$$\boxed{M_b^2(\Phi_0) = (m^2 + \Gamma \,\mathrm{tr}(\Phi_0^2)) I_3 + \kappa \Phi_0.} \tag{57}$$

If a_1, a_2, a_3 are the real eigenvalues of Φ_0, the three complex transverse eigenvalues are

$$m^2 + \Gamma \,\mathrm{tr}(\Phi_0^2) + \kappa a_i. \tag{58}$$

A negative value gives an explicit classical transverse instability away from the old S^8 sector.

Proof. Every term is invariant under $\Phi \mapsto U \Phi U^*$, $b \mapsto U b$, and the determinant identity (17). All scalar fields have engineering dimension one; the cubic coefficient κ has dimension one and the quartic coefficient Γ is dimensionless. The positive quartic term dominates the cubic invariant, so the potential is bounded below.

Expand the potential to second order in b at $b = 0$. The radial terms give $(m^2 + \Gamma \,\mathrm{tr}(\Phi_0^2)) b^* b$, while $-\kappa \mathcal{N}_{14}$ in the potential contributes $\kappa b^* \Phi_0 b$. Diagonalizing Φ_0 gives (58). □

7.3 Exact vacuum phase diagram and the unlocked internal coordinate

For $Q = \Theta(\Phi, b) \in \mathrm{Herm}_3^0(\mathbb{H})$ put

$$s(Q) = \mathrm{tr}(Q^2) = \mathrm{tr}(\Phi^2) + 2b^* b, \qquad n(Q) = \det_{\mathbb{H}} Q = \det \Phi - b^* \Phi b. \tag{59}$$

The scalar potential in (56) is

$$V_{14}(Q) = \frac{m^2}{2} s(Q) + \frac{\Gamma}{4} s(Q)^2 - \kappa n(Q). \tag{60}$$

Although only the subgroup SU(3) is gauged, this potential is invariant under the full compact Jordan automorphism group Sp(3).

Lemma 7.4 (Sharp trace-free determinant bound)**.** *For every* $Q \in \mathrm{Herm}_3^0(\mathbb{H})$*,*

$$\boxed{54\, n(Q)^2 \le s(Q)^3.} \tag{61}$$

For $Q \neq 0$*, equality holds precisely when*

$$Q = x(3P - I_3) \tag{62}$$

for a nonzero $x \in \mathbb{R}$ *and a primitive quaternionic projector* $P \in \mathcal{P}_{\mathbb{H}} \cong \mathbb{HP}^2$*. In that case*

$$s(Q) = 6x^2, \qquad n(Q) = 2x^3.$$

Proof. Quaternionic Hermitian spectral theory gives real eigenvalues x_1, x_2, x_3 with $x_1 + x_2 + x_3 = 0$. At fixed $s = x_1^2 + x_2^2 + x_3^2$, Lagrange multipliers for $x_1 x_2 x_3$ force two eigenvalues to coincide at a nonzero extremum. Thus the extremal spectrum is a permutation of $(2x, -x, -x)$. Substitution gives $n^2 = 4x^6 = s^3/54$. Equality matrices are exactly the $\mathrm{Sp}(3)$ conjugates of $x\,\mathrm{diag}(2, -1, -1) = x(3E_{11} - I_3)$, which are (62). □

Theorem 7.5 (Complete homogeneous vacuum diagram)**.** *Assume* $\Gamma > 0$*,* $m^2 \ge 0$*, and* $\kappa \neq 0$*. Define*

$$m_{\mathrm{c}}^2 = \frac{\kappa^2}{27\Gamma}. \tag{63}$$

Then the global minima of (60) *are exactly as follows.*

1. *If* $m^2 > m_{\mathrm{c}}^2$*, the unique minimum is* $Q = 0$*.*
2. *If* $m^2 = m_{\mathrm{c}}^2$*, the minima are* $Q = 0$ *together with*

$$Q = \frac{\kappa}{9\Gamma}(3P - I_3), \qquad P \in \mathbb{HP}^2. \tag{64}$$

3. *If* $0 \le m^2 < m_{\mathrm{c}}^2$*, the minimum set is the single* $\mathrm{Sp}(3)$ *orbit*

$$\boxed{\mathcal{V}_{\min} = \{x_*(3P - I_3) : P \in \mathbb{HP}^2\} \cong \mathbb{HP}^2,} \tag{65}$$

where

$$x_* = \frac{\kappa + \mathrm{sgn}(\kappa)\sqrt{\kappa^2 - 24\Gamma m^2}}{12\Gamma}. \tag{66}$$

For $\kappa = 0$*, the origin is the unique global minimum.*

At every nonzero minimum, the scalar Hessian with respect to the metric $\mathrm{tr}(\delta Q^2)$ *has eight zero eigenvalues tangent to* $\mathbb{HP}^2$*, one radial eigenvalue*

$$M_{\mathrm{rad}}^2 = x_*(12\Gamma x_* - \kappa) > 0, \tag{67}$$

and a fivefold normal eigenvalue

$$M_5^2 = 3\kappa x_* > 0. \tag{68}$$

After gauging $\mathrm{SU}(3)$*, the vacuum orbit space is the interval in* (25)*. The unbroken gauge groups are respectively* $S(U(1) \times U(2))$*,* $U(1)_{1,-1,0}$*, and* $\mathrm{SU}(2)$ *at* $\delta = 0$*,* $0 < \delta < 1$*, and* $\delta = 1$*. A generic vacuum therefore has one exact gauge-invariant flat internal coordinate.*

Proof. Let $r = \sqrt{s(Q)}$. By Theorem 7.4, minimization at fixed r chooses

$$n(Q) = \operatorname{sgn}(\kappa)\frac{r^3}{3\sqrt{6}},$$

and equality forces (62). The remaining radial problem is

$$f(r) = \frac{m^2}{2}r^2 + \frac{\Gamma}{4}r^4 - \frac{|\kappa|}{3\sqrt{6}}r^3, \qquad r \geq 0.$$

Put $a = |\kappa|/\sqrt{6}$. Its nonzero stationary points solve $\Gamma r^2 - ar + m^2 = 0$. At such a point,

$$f(r) = r^3\left(\frac{a}{6} - \frac{\Gamma r}{4}\right).$$

The larger stationary point has negative value precisely when $m^2 < \kappa^2/(27\Gamma)$ and zero value at equality. Converting $r = \sqrt{6}|x|$ gives (64)–(66).

For the Hessian, fix $P = E_{11}$, so $Q_* = x_* \operatorname{diag}(2, -1, -1)$. Tangent directions to the Sp(3) orbit have zero Hessian by invariance. The radial calculation gives (67). The remaining five-dimensional normal representation is a traceless quaternionic Hermitian 2×2 block B_0. To quadratic order,

$$s(Q_* + B_0) = 6x_*^2 + \operatorname{tr}(B_0^2), \qquad n(Q_* + B_0) = 2x_*^3 - x_* \operatorname{tr}(B_0^2).$$

Using the stationary equation $m^2 - \kappa x_* + 6\Gamma x_*^2 = 0$ gives

$$V_{14}(Q_* + B_0) - V_{14}(Q_*) = \frac{1}{2}(3\kappa x_*)\operatorname{tr}(B_0^2) + O(\|B_0\|^3),$$

which proves (68). The gauge stabilizers and the interval quotient are those of Theorem 4.5. □

Corollary 7.6 (Exact unlock/relock control)**.** *On the nonzero vacuum manifold* (65)*, the additional renormalizable* SU(3)*-invariant perturbation*

$$\Delta V_\nu = \nu b^* b \tag{69}$$

restricts exactly to

$$\boxed{\Delta V_\nu = \frac{9\nu x_*^2}{4}\,\delta.} \tag{70}$$

Thus, among the formerly degenerate vacua, $\nu > 0$ selects the old complex endpoint $\mathbb{CP}^2$ at $\delta = 0$, $\nu < 0$ selects the quaternionic endpoint S^5 at $\delta = 1$, and $\nu = 0$ leaves the full internal interval unlocked. The exact morphism relating this control parameter to the topology is $Q \mapsto b^*b \mapsto \delta$ from (29)*.*

Remark 7.7 (Why fourteen rather than thirteen for full colour). The condition $u^*b = 0$ defining the Lorentz slice $\mathcal{L}_{13}$ chooses one colour line and is not invariant under all of SU(3). Full local colour invariance retains the whole triplet $b \in \mathbb{C}^3$, giving fourteen trace-free scalar components or fifteen components when the trace singlet of A is restored. Thus 13 is the exact symmetry-reduced $4 + 4 + 4 + 1$ model; 14 is the minimal linear SU(3) carrier that unlocks the old S^8; and 15 is the full positive Jordan cone.

8 A rigorous finite-lattice quantum dynamics

The classical colour model can be quantized on a finite spatial lattice without replacing the compact link variables by Lie-algebra coordinates. Let $\Lambda = (V, E, \mathcal{P})$ be a finite connected oriented lattice with a finite set of oriented plaquettes. Put

$$\mathcal{S}_{14} = \mathrm{Herm}_3^0(\mathbb{C}) \oplus \mathbb{C}^3$$

with Euclidean norm

$$\|(\Phi, b)\|^2 = \mathrm{tr}(\Phi^2) + 2b^*b,$$

and let

$$\rho(U)(\Phi, b) = (U\Phi U^*, Ub).$$

The finite configuration manifold is

$$\mathfrak{M}_\Lambda = \mathrm{SU}(3)^E \times \mathcal{S}_{14}^V. \tag{71}$$

If e is directed from $s(e)$ to $t(e)$, use the gauge convention

$$U_e \longmapsto g_{t(e)} U_e g_{s(e)}^{-1}, \qquad Q_v \longmapsto \rho(g_v) Q_v, \qquad g \in \mathrm{SU}(3)^V. \tag{72}$$

Fix positive constants

$$a_g, a_s, \beta_p, \beta_h > 0.$$

For a plaquette p, let U_p be its ordered holonomy. Define the real potential

$$\begin{aligned} W_\Lambda(U, Q) = {} & \beta_p \sum_{p \in \mathcal{P}} (3 - \mathrm{Re}\,\mathrm{tr}\, U_p) + \beta_h \sum_{e \in E} \|Q_{t(e)} - \rho(U_e) Q_{s(e)}\|^2 \\ & + \sum_{v \in V} V_{14}(Q_v). \end{aligned} \tag{73}$$

No linearization of U_e and no deletion of the b-triplet occurs.

Theorem 8.1 (Finite-lattice Hamiltonian and exact Gauss law)**.** *On*

$$\mathscr{H}_\Lambda = L^2\left(\mathrm{SU}(3)^E \times \mathcal{S}_{14}^V, \prod_{e \in E} dU_e \prod_{v \in V} dQ_v\right), \tag{74}$$

where each dU_e is normalized Haar measure and each dQ_v is Lebesgue measure for the displayed Euclidean norm, consider

$$H_\Lambda = -a_g \sum_{e \in E} \Delta_{\mathrm{SU}(3),e} - a_s \sum_{v \in V} \Delta_{\mathcal{S},v} + W_\Lambda. \tag{75}$$

Initially define it on $C_c^\infty(\mathfrak{M}_\Lambda)$. Then:

1. *H_Λ is bounded below and its quadratic form has a canonical self-adjoint Friedrichs realization, with $C_c^\infty(\mathfrak{M}_\Lambda)$ as a form core;*

2. *its resolvent is compact, so its spectrum is a discrete sequence*

$$E_0 < E_1 \le E_2 \le \cdots, \qquad E_n \to \infty;$$

3. *the unitary representation* $\mathcal{U}_\Lambda$ *induced by* (72) *commutes with* H_Λ*, and the exact Gauss-law projection is*

$$\boxed{P_{\text{phys}} = \int_{\mathrm{SU}(3)^V} \mathcal{U}_\Lambda(g)\, dg;} \tag{76}$$

4. *the positive ground-state eigenfunction of* H_Λ *is unique and lies in*

$$\mathscr{H}_{\Lambda,\text{phys}} = P_{\text{phys}} \mathscr{H}_\Lambda;$$

5. *the reduced operator*

$$H_{\Lambda,\text{phys}} = H_\Lambda\big|_{\mathscr{H}_{\Lambda,\text{phys}}} \tag{77}$$

has compact resolvent, a one-dimensional ground state, and a strictly positive finite-volume excitation gap

$$\boxed{\Delta_\Lambda = \inf\big(\mathrm{Spec}(H_{\Lambda,\text{phys}}) \setminus \{E_0\}\big) - E_0 > 0.} \tag{78}$$

Proof. The plaquette and hopping terms are nonnegative. By the sharp determinant bound (61), each local potential satisfies

$$V_{14}(Q) \geq \frac{\Gamma}{4}\|Q\|^4 - \frac{|\kappa|}{3\sqrt{6}}\|Q\|^3,$$

so W_Λ is bounded below and tends to $+\infty$ whenever any scalar coordinate tends to infinity, uniformly in the compact link variables. The associated semibounded quadratic form is closable; its closure defines the Friedrichs extension.

The configuration manifold is a finite product of compact Lie groups and Euclidean spaces, and the potential is confining in every Euclidean direction. The form-domain embedding into L^2 is therefore compact: truncate to a large scalar ball, use Rellich compactness on the resulting compact manifold with boundary, and use coercivity to make the exterior L^2 norm uniformly small. This proves compact resolvent.

Every term in (75) is invariant under (72). Haar averaging therefore gives the orthogonal projection (76), which strongly commutes with the Hamiltonian. The scalar Schrödinger semigroup on the connected configuration manifold is positivity improving. Hence the lowest eigenvalue is simple and has a strictly positive eigenfunction. Gauge invariance and uniqueness force that eigenfunction to be fixed by $\mathrm{SU}(3)^V$. Restriction to the reducing physical subspace preserves compact resolvent and the simple ground state; discreteness then gives (78). □

Remark 8.2 (What this quantum theorem decides). The Hamiltonian (75) is one fixed nonlinear compact-link quantum system, rather than a Gaussian Hessian on a Lie-algebra chart. It has an exact Gauss-law projection and retains the fields that unlock the old S^8. The theorem gives a positive gap for each fixed finite lattice. It does not assert a uniform lower bound as lattice spacing tends to zero or spatial volume tends to infinity. Those limits define the next operator problem; a finite-volume gap alone decides neither confinement nor the pure four-dimensional Yang–Mills mass gap.

9 Exact status

The following objects are now constructed with explicit maps.

1. A thirteen-dimensional triality cone $\mathcal{T}_{13}$ with a cubic three-edge interaction, cyclic symmetry, compactified boundary S^{12}, and globally well-posed nonlinear mechanics.

2. A division-algebra ladder. Its equal-diagonal determinant boundaries are S^3, S^6, S^{12}, and S^{24}.

3. The fifteen-dimensional quaternionic Jordan closure, including the two diagonal anisotropy modes generated by Jordan multiplication and determinant dynamics.

4. A fourteen-dimensional SU(3) representation $\mathrm{Herm}_3^0(\mathbb{C}) \oplus \mathbb{C}^3$ in which the old S^8 embeds with six explicit transverse modes; full colour forces dimension fourteen (or fifteen after restoring trace).

5. A proper positive projection from the quaternionic cone to the complex cone, with exact $B^6/B^2/\mathrm{pt}$ fibres, and a commuting Gram-boundary square $S^8 \hookrightarrow S^{14}$.

6. The cohomogeneity-one topology $\mathbb{HP}^2/\mathrm{SU}(3) = [0,1]$, with endpoint orbits $\mathbb{CP}^2$ and S^5, together with its double disk-bundle decomposition.

7. A local $\mathrm{SU}(2)^3$ triangle gauge theory and a local SU(3) Yang–Mills–scalar theory, including the complete homogeneous vacuum phase diagram, Hessian, unbroken subgroups, and an exact unlock/relock perturbation.

8. A nonlinear finite-lattice quantum Hamiltonian with compact SU(3) links, all fourteen scalar modes, product Haar–Lebesgue measure, an exact vertex Gauss-law projection, compact resolvent, a unique physical ground state, and a positive gap on every fixed finite lattice.

These constructions do not identify the equal-diagonal complex S^6 with a separately proposed torus-fibred complex structure on S^6. They also do not construct pure quantum Yang–Mills: both local actions contain scalar fields, and the finite-lattice Hamiltonian has no proved reflection-positive transfer-matrix reconstruction, renormalized continuum limit, infinite-volume limit, or uniform pure-gauge mass gap.

The next exact calculations are now sharply defined. For the quantum model one must choose a lattice-spacing/volume scaling of all displayed couplings and determine whether Δ_Λ has a nonzero, zero, or divergent limit after vacuum-energy subtraction. For the geometry one must test whether the toroidal monodromy and filling cocycles reduce to the explicit $\mathrm{Sp}(1)^3 times C_3$, SU(3), or Sp(3) structure groups above. That test requires the actual overlap matrices and boundary inclusions, not a sphere-dimension match.

References

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